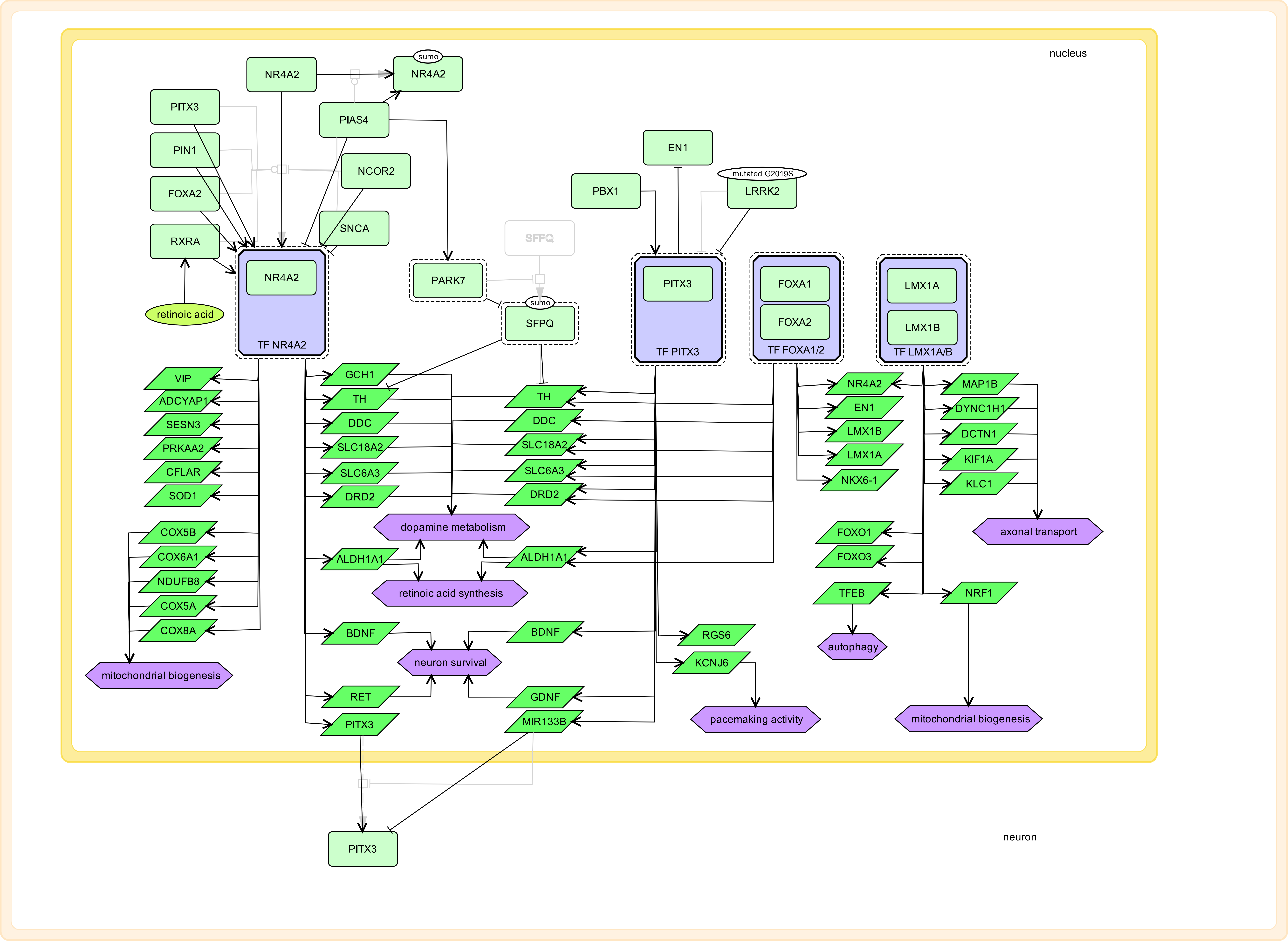
nucleus
neuron
NR4A2
sumo
NR4A2
PITX3
PIAS4
PIN1
NCOR2
FOXA2
SNCA
RXRA
NR4A2
TF NR4A2
retinoic acid
PARK7
SFPQ
sumo
SFPQ
EN1
PBX1
mutated G2019S
LRRK2
PITX3
TF PITX3
FOXA1
FOXA2
TF FOXA1/2
LMX1A
LMX1B
TF LMX1A/B
VIP
ADCYAP1
SESN3
PRKAA2
CFLAR
SOD1
COX5B
COX6A1
NDUFB8
COX5A
COX8A
mitochondrial biogenesis
GCH1
TH
DDC
SLC18A2
SLC6A3
DRD2
dopamine metabolism
ALDH1A1
retinoic acid synthesis
BDNF
neuron survival
RET
PITX3
TH
DDC
SLC18A2
SLC6A3
DRD2
ALDH1A1
BDNF
GDNF
MIR133B
RGS6
KCNJ6
pacemaking activity
NR4A2
EN1
LMX1B
LMX1A
NKX6-1
FOXO1
FOXO3
TFEB
autophagy
MAP1B
DYNC1H1
DCTN1
KIF1A
KLC1
axonal transport
NRF1
mitochondrial biogenesis
PITX3

retinoic acid
NR4A2
PIAS4
RXRA
SNCA
NCOR2
PITX3
PIN1
sumo
NR4A2
FOXA2
PARK7
PBX1
mutated G2019S
LRRK2
LMX1A
LMX1B
TF LMX1A/B
nucleus
NR4A2
TF NR4A2
FOXA1
FOXA2
TF FOXA1/2
sumo
SFPQ
PITX3
TF PITX3
FOXO1
DCTN1
DYNC1H1
KIF1A
KLC1
MAP1B
TFEB
NRF1
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PITX3